Incidence representations: complete measurement inputs

Human Compression Project

This is an input proposal for the accepted two-source scientific edition. It evaluates no compression factor. Its primary coordinate remains **calibrated selected text, NFC, characters without whitespace**, exactly as in the earlier F010 comparison. Independent input acceptance is still required.

Every source and every complete paper

F010 is Díaz Ramos–Molinier–Viruel, *Every finite group is represented by a finite incidence geometry*, arXiv:2511.02410v2 (16 December 2025). L126 is Leemans–Stokes–Tranchida, *Groups represented by incidence geometries*, arXiv:2506.10464v1 (12 June 2025; manuscript dated 13 June). Both exact 13-page originals are included, with their titles, authors, references and arXiv identification.

Complete document	Pages	Raw NFC nonspace chars	Primary NFC nonspace chars
F010 original	13	25,950	25,348
L126 original	13	35,651	35,124
Complete shared theory	13	36,732	36,715
Complete F010 application	4	10,989	10,985
Complete L126 application	12	34,467	34,452
Separate full program appendix	16	35,746	35,723

The written boundary pays the whole 13-page common paper and both whole applications, including the final reference pages: $13 + 4 + 12 = 29$ replacement pages. The separate appendix has all 16 pages. With the originals this is 71 physical pages, of which 55 are written source/common/application pages. There is no discount for inherited proofs, repeated context or bibliography.

The finite-normal-pair encoder and its full proof now belong to the common paper. Both applications invoke that shared theorem. Neither has an unpaid sibling-application premise. A later F010 conditional comparison must first acquire the whole common paper and actual retained classical inputs. A later L126 conditional comparison must also acquire its complete required graph certificate and graph data. Optional validation is not a proof prerequisite.

What is unchanged and what is re-extracted

Every one of the 30 accepted F010 *source* coordinates is preserved with its exact font, glyph and furniture evidence. All other current PDFs are extracted anew, including the changed four-page F010 application. No old application, common-paper or appendix total is substituted.

For every PDF the packet retains three variants (literal raw, furniture-only selected, glyph-calibrated selected), NFC and NFKC, and five units: whitespace tokens, alphanumeric units, nonspace characters, collapsed-whitespace characters, and collapsed-whitespace UTF-8 bytes. The resulting 180 PDF cells include 30 inherited cells. The 17 whole raw files supply another 170 cells.

Symmetric programs and retained costs

Neither original nor either current scientific application/common paper contains a concrete executable or pseudocode listing. All manuscript listing masks are therefore explicitly empty. The source's Tits construction and the new written algorithms are mathematical statements and proofs, so they remain counted. Programs are excluded from both sides of manuscript comparisons.

The separate appendix contains ten whole mathematical files: one required finite certificate and nine optional validations. The required certificate uses the entire 19,326-byte graph JSON. The appendix does not replace that file with a summary. Printed and raw versions of the same program must never be added together. All whole-file costs below are an inventory, not a software-inclusive ratio.

Whole raw category	Files	Bytes	NFC nonspace
Required finite certificate	1	8,030	6,070
Mandatory complete graph data	1	19,326	19,315
Optional validation programs	9	35,601	28,753
Operational helpers	2	16,622	15,111
Partial Lean evidence	4	15,931	11,706

The two helpers orchestrate scientific and migration checks. The four Lean files contain 22 scoped declarations, not a formal proof of either entire source or of the concrete graph. The mandatory pure-Python certificate establishes the one explicitly given 896-vertex graph; optional GAP, nauty or Bliss checks are not its proof premises. All source software, retained external-proof costs, library costs and full graphic structure remain unknown. Thus no software-inclusive factor is available and missing cost is not zero.

Exact extraction and visible symbols

The parser versions are pypdf 6.13.0 and FontTools 4.63.0. Selected L126 text removes only the exact running headers/page numbers; its arXiv banner remains. Own selected PDFs remove only physical page numbers. Every page and reference line is otherwise retained. The F010 selection and calibration are unchanged, including the visible right braces encoded as tab characters.

The new L126 audit binds 24 actual font dictionaries and 649 encoded classes, using dictionary identity, effective encoding and actual embedded outlines. Eight classes at 28 occurrences calibrate script and blackboard capitals and the left component of a mapsto symbol. The adjacent arrow is kept separately. Fifty-eight exceptional source and replacement outlines were directly viewed. Phi variants and ligatures are retained as actually extracted. The current own PDFs receive no glyph substitutions. Source diagrams on pages 2, 4, 5 and 12 retain their extracted labels; their complete graphical cost is unknown.

Scope and independence

The accepted science preserves six F010 principal blocks and 31 L126 dispositions, including explicit corrections and retained classical inputs. The finite-field repair, free-basis qualification and chamber-connectivity condition remain visible. The broader Section 6.2 and Section 8 programmes are not declared solved. No new question, priority or formalization credit is claimed. This packet reuses distinct scientific acceptance; its author also authored the current L126 proof and migration. A separate reviewer must decide whether these proposed inputs are suitable for numerical comparison.